

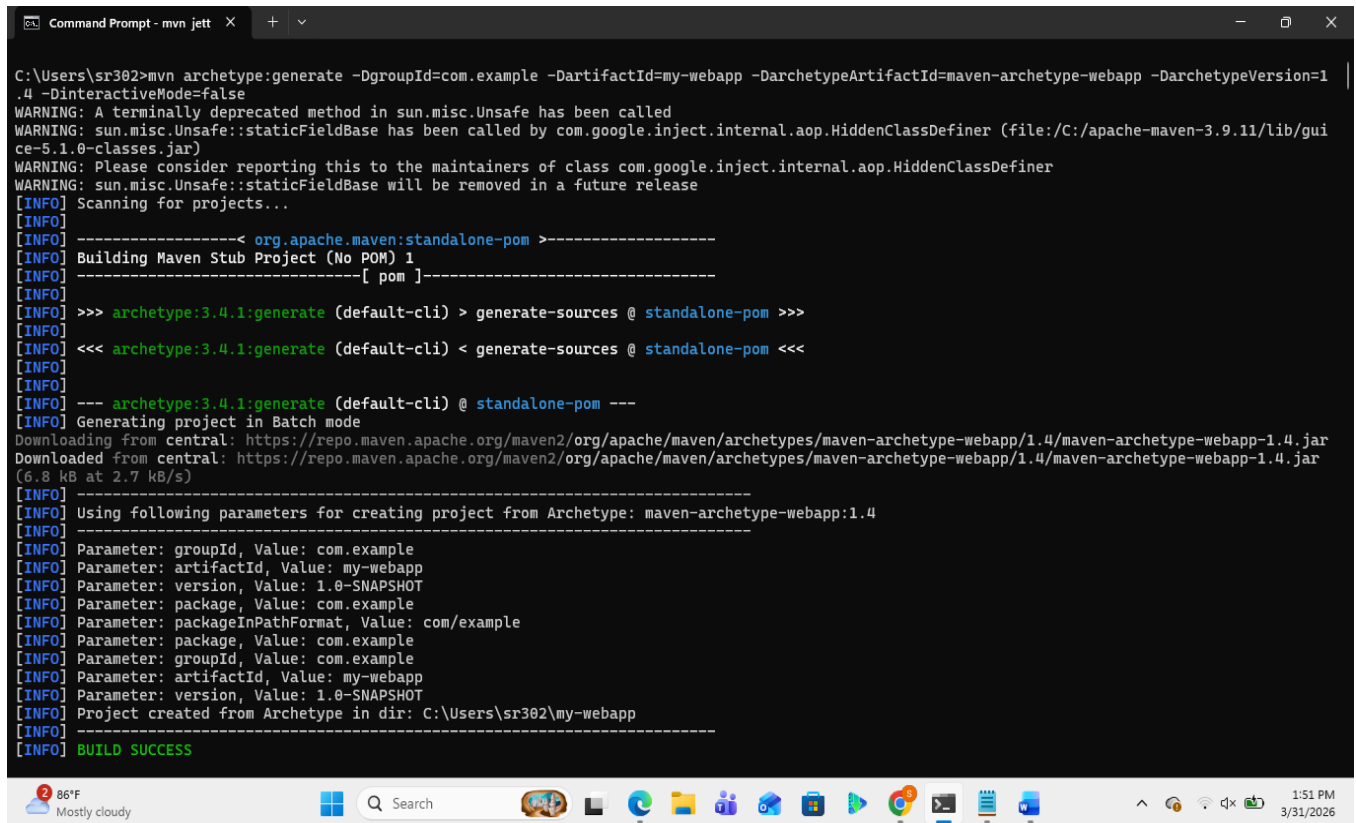
INT-332 Practical-4

Maven Practical

Exercise 1: Maven jetty.

Step 1: Create Maven Project

`mvn archetype:generate -DgroupId=com.example -DartifactId=my-webapp -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false`



```
C:\Users\sr302>mvn archetype:generate -DgroupId=com.example -DartifactId=my-webapp -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/gui
ce-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/maven-archetype-webapp-1.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/maven-archetype-webapp-1.4.jar
(6.8 kB at 2.7 kB/s)
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] -----
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-webapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: packageInPathFormat, Value: com/example
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-webapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Archetype in dir: C:\Users\sr302\my-webapp
[INFO] -----
[INFO] BUILD SUCCESS
```

Step 2: Package Application:-

Mvn clean package

```
Command Prompt - mvn jett x + v
C:\Users\sr302\my-webapp>mvn clean package
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassD
efiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:my-webapp >-----
[INFO] Building my-webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ my-webapp ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ my-webapp ---
[INFO] skip non existing resourceDirectory C:\Users\sr302\my-webapp\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ my-webapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ my-webapp ---
[INFO] skip non existing resourceDirectory C:\Users\sr302\my-webapp\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ my-webapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ my-webapp ---
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ my-webapp ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.4.0:war (default-war) @ my-webapp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [my-webapp] in [C:\Users\sr302\my-webapp\target\my-webapp-1.0-SNAPSHOT]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\Users\sr302\my-webapp\src\main\webapp]
[INFO] Building war: C:\Users\sr302\my-webapp\target\my-webapp-1.0-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.144 s
[INFO] Finished at: 2026-03-31T12:41:15+05:30
```

Step 3: Modify index.jsp Program:-

With path : src/main/java/com/example/index.jsp

```
index.jsp pom.xml +
File Edit View
<html>
<body>
<h2>Hello! This webpage is running using Maven + Jetty.</h2>
</body>
</html>
```

Step 4: Add dockerfile-maven-plugin in pom.xml:-

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>my-webapp</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>war</packaging>

  <name>my-webapp</name>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
  </dependencies>

  <build>
    <plugins>

    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
```

```
<artifactId>maven-compiler-plugin</artifactId>

<version>3.11.0</version>

<configuration>

  <source>${maven.compiler.source}</source>

  <target>${maven.compiler.target}</target>

</configuration>

</plugin>


<plugin>

  <groupId>org.eclipse.jetty</groupId>

  <artifactId>jetty-maven-plugin</artifactId>

  <version>9.4.54.v20240208</version>


  <configuration>

    <scanIntervalSeconds>3</scanIntervalSeconds>

    <httpConnector>

      <port>8081</port>

    </httpConnector>

    <webApp>

      <contextPath>/my-webapp</contextPath>

    </webApp>

  </configuration>

</plugin>

</plugins>

</build>

</project>
```

Step 5: Run Docker Container:-

mvn jetty:run-war

```
Command Prompt - mvn jett X + v
C:\Users\sr302\my-webapp>mvn jetty:run-war
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-maven-plugin/9.4.54.v20240208/jetty-maven-plugin-9.4.54.v20240208.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-maven-plugin/9.4.54.v20240208/jetty-maven-plugin-9.4.54.v20240208.pom (12 kB at 4.7 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-project/9.4.54.v20240208/jetty-project-9.4.54.v20240208.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-project/9.4.54.v20240208/jetty-project-9.4.54.v20240208.pom (73 kB at 620 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/testcontainers/testcontainers-bom/1.16.1/testcontainers-bom-1.16.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/testcontainers/testcontainers-bom/1.16.1/testcontainers-bom-1.16.1.pom (7.2 kB at 22 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/infinispan/infinispan-bom/11.0.18.Final/infinispan-bom-11.0.18.Final.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/infinispan/infinispan-bom/11.0.18.Final/infinispan-bom-11.0.18.Final.pom (18 kB at 55 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/infinispan/infinispan-build-configuration-parent/11.0.18.Final/infinispan-build-configuration-parent-11.0.18.Final.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/infinispan/infinispan-build-configuration-parent/11.0.18.Final/infinispan-build-configuration-parent-11.0.18.Final.pom (16 kB at 45 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/jboss/jboss-parent/36/jboss-parent-36.pom
```

```
Command Prompt - mvn jett X + v
[INFO] -----< com.example:my-webapp >-----
[INFO] Building my-webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] >>> jetty:9.4.54.v20240208:run-war (default-cli) > package @ my-webapp >>>
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ my-webapp ---
[INFO] skip non existing resourceDirectory C:\Users\sr302\my-webapp\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ my-webapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ my-webapp ---
[INFO] skip non existing resourceDirectory C:\Users\sr302\my-webapp\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ my-webapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ my-webapp ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.4.0:war (default-war) @ my-webapp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [my-webapp] in [C:\Users\sr302\my-webapp\target\my-webapp-1.0-SNAPSHOT]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\Users\sr302\my-webapp\src\main\webapp]
[INFO] Building war: C:\Users\sr302\my-webapp\target\my-webapp-1.0-SNAPSHOT.war
```

```
Command Prompt - mvn jett X + v
Downloaded from central: https://repo.maven.apache.org/maven2/javax/transaction/javax.transaction-api/1.3/javax.transaction-api-1.3.jar (28 kB at 789 B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/ben-manes/caffeine/caffeine/2.9.3/caffeine-2.9.3.jar (912 kB at 25 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/mortbay/jasper/apache-jsp/8.5.70/apache-jsp-8.5.70.jar (1.1 MB at 29 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/taglibs/taglibs-standard-impl/1.2.5/taglibs-standard-impl-1.2.5.jar (206 kB at 5.6 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jdt/ecj/3.19.0/ecj-3.19.0.jar (2.9 MB at 66 kB/s)
[INFO] Logging initialized @80869ms to org.eclipse.jetty.util.log.Slf4jLog
[INFO] Configuring Jetty for project: my-webapp
[INFO] Context path = /my-webapp
[INFO] Tmp directory = C:\Users\sr302\my-webapp\target\tmp
[INFO] Web defaults = org/eclipse/jetty/webapp/webdefault.xml
[INFO] Web overrides = none
[INFO] jetty-9.4.54.v20240208; built: 2024-02-08T19:42:39.027Z; git: cef3fbd6d736a21e7d541a5db490381d95a2047d; jvm 25.0.1+8-LTS-27
[INFO] Scanning elapsed time=61ms
[INFO] DefaultSessionIdManager workerName=node0
[INFO] No SessionScavenger set, using defaults
[INFO] node0 Scavenging every 600000ms
[INFO] Started o.e.j.m.p.JettyWebAppContext@3f9f8d23{Archetype Created Web Application,/my-webapp,file:///C:/Users/sr302/my-webapp/target/my-webapp-1.0-SNAPSHOT/,AVAILABLE}{C:\Users\sr302\my-webapp\target\my-webapp-1.0-SNAPSHOT.war}
[INFO] Started ServerConnector@632b5c79{HTTP/1.1, (http/1.1)}{0.0.0.0:8081}
[INFO] Started @82451ms
[INFO] Started Jetty Server
```

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